

PEACH-Golden: Enhanced Localization Efficiency and Accuracy via Golden Section Search in MUSIC

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Abstract—This work presents PEACH-Golden, a two-stage refinement of the PEACH-MUSIC (Position Estimation Algorithm Using Circles and Hyperbolas in MUSIC) [1] localization method. Renowned for its efficient geometry and low computational complexity, PEACH-MUSIC is first employed to generate an initial, potentially coarse, position estimate. Subsequently, PEACH-Golden refines this estimate via Golden Section Search (GSS), reducing objective function evaluations for precision. This allows the initial PEACH stage to be computationally lighter. Numerical results show PEACH-Golden achieves improved accuracy over standard PEACH-MUSIC implementations with a substantially lower execution time, enhancing overall performance.

Keywords—Position Estimation Algorithm using Circle and Hyperbola (PEACH), user localization, Cramér-Rao Bound (CRB), Uniform Rectangular Array (URA), Golden section search (GSS).

I. INTRODUCTION

Managing the trade-off is necessary to address scenarios where computational complexity is critical. In far-field scenarios with high URA, focus formation over the user region in the pseudospectrum MUSIC becomes feasible [2]. In such cases, PEACH-MUSIC [1] is a method that stands out, reducing the complexity of the Multiple Signal Classification (MUSIC) algorithm [3] [4]. We define as optimal the configuration in which the number of candidates reaches saturation in the PEACH performance. In this work, we propose the PEACH-Golden method, which performs two-step refinement to PEACH-MUSIC, including the GSS method [5]. The goal is to achieve optimal PEACH performance with a significant reduction in complexity. CRB is adopted for aggregation in the performance evaluation, which provides the minimum variance that can be obtained, given a physical scenario. Nelder-Mead is the heuristic method [6] that will be used as a refinement for comparison.

II. THE PEACH LOCALIZATION METHODOLOGY

The challenge of accurate and computationally efficient user localization, particularly with elevated antenna arrays common in urban ISAC scenarios, in this work is addressed by the PEACH-MUSIC framework [1]. The core PEACH-MUSIC method leverages a novel geometric interpretation of the MUSIC pseudo-spectrum. It uniquely decomposes a URA into its constituent horizontal and vertical Uniform Linear Arrays (ULAs). This decomposition reveals distinct spectral patterns: the horizontal ULA generates a hyperbolic detection region, while the vertical ULA produces a circular region in

the user's plane. The PEACH-MUSIC algorithm then analytically computes the user's 2D position by determining the intersection of these geometric shapes, offering a closed-form solution that significantly reduces the computational burden compared to traditional multi-dimensional grid searches of the pseudo-spectrum.

To further enhance localization accuracy and robustness, particularly in environments with lower SNR, the PEACH-MUSIC-NM method was also proposed in [1]. This two-stage approach first utilizes the computationally efficient PEACH-MUSIC algorithm to obtain an initial coarse position estimate. Subsequently, a Nelder-Mead (NM) optimization algorithm performs a refined local search around this initial estimate, leveraging the full URA pseudo-spectrum to converge towards a more precise 3D user location. The NM stage, being a gradient-free simplex method, adds minimal computational overhead while improving the system's resilience to noise. As demonstrated in, the PEACH-MUSIC framework, especially with the NM refinement, achieves near-optimal localization accuracy, approaching the CRB, while maintaining significantly lower computational complexity than conventional MUSIC-based grid search techniques.

III. THEORETICAL PERFORMANCE BOUND

To benchmark the proposed localization algorithm, we utilize the Cramér-Rao Bound (CRB) derived for 2D user position estimation ($\mathbf{u} = [x, y]^T$) in the context of elevated planar antenna arrays. The CRB provides the minimum achievable variance for any unbiased estimator. For the coordinates x and y , the CRB is given by [1]:

$$\text{CRB}(x) = \left[J_{xx} - \frac{J_{xy}^2}{J_{yy}} \right]^{-1}, \quad \text{CRB}(y) = \left[J_{yy} - \frac{J_{xy}^2}{J_{xx}} \right]^{-1}. \quad (1)$$

The elements of the Fisher Information Matrix (FIM), J_{xx} , J_{yy} , and J_{xy} , as well as the normalized power factor $P_{\text{norm}} = L(|s|\lambda/(\sqrt{2}\pi\sigma_n))^2$, are defined based on the specific array geometry, user location, number of snapshots L , signal amplitude $|s|$, wavelength λ , and noise variance σ_n^2 . For their detailed expressions and derivation in the elevated array scenario, readers are referred to [1]. This CRB formulation accounts for path-loss amplitude and phase information inherent in the signal model.

IV. PEACH-GOLDEN

The Golden Section Search (GSS) method facilitates the estimation of a user's xy -position by optimizing one-dimensional functions derived from the MUSIC pseudo-spectrum [7]. However, it requires that the MUSIC pseudo-spectrum functions—associated with the horizontal ULA,

$P_h(\cdot)$, and the vertical ULA, $P_v(\cdot)$ exhibit unimodal behavior within the defined search interval. This unimodality requirement makes it crucial to mitigate sidelobes and spurious peaks, often induced by noise. The PEACH-Golden method addresses this by limiting the search intervals for GSS within the neighborhood (ξ) of the estimated distance difference between the ULA endpoints (x_0) and the radial distance projected onto the horizontal plane (y_0), obtained by PEACH-MUSIC. This localized search strategy significantly increases the probability of encountering a single, well-defined peak within each GSS interval $[a, b]$, thereby ensuring the effectiveness of the GSS refinement.

GSS employs the Golden Ratio (ϕ) to determine internal points for interval reduction strategically. The method iteratively narrows the search interval by evaluating the relevant MUSIC pseudo-spectrum function (e.g., $P_h(\cdot)$ or $P_v(\cdot)$), requiring typically only one new function evaluation per iteration after the initial setup. This allows for efficient convergence towards the optimal coordinate (x_0^* or y_0^*) for the optimized dimension. Algorithm 1 presents this GSS-based refinement process, which is applied within the neighborhood defined by the initial coarse estimate from PEACH-MUSIC.

Algorithm 1 PEACH-Golden

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1: Input:  $x_0, y_0, P_h, P_v, \xi, \epsilon$ 
2: Initialize  $\phi = (\sqrt{5} - 1)/2$ 
3: function GOLDEN( $P_{\text{MUSIC}}, a, b, \epsilon$ )
4:    $c_1 = b - \phi(b - a), c_2 = a + \phi(b - a)$ 
5:    $f_1 = P_{\text{MUSIC}}(c_1), f_2 = P_{\text{MUSIC}}(c_2)$ 
6:   while  $b - a > \epsilon$  do
7:     if  $f_1 > f_2$  then
8:        $a = c_1, c_1 = c_2, f_1 = f_2$ 
9:        $c_2 = a + \phi(b - a), f_2 = P_{\text{MUSIC}}(c_2)$ 
10:    else
11:       $b = c_2, c_2 = c_1, f_2 = f_1$ 
12:       $c_1 = b - \phi(b - a), f_1 = P_{\text{MUSIC}}(c_1)$ 
13:    end if
14:  end while
15:  return  $\arg \max\{f(c_1), f(c_2)\}$ 
16: end function
17:  $x = \text{GOLDEN}(P_h, x_0 - \xi, x_0 + \xi, \epsilon)$ 
18:  $y = \text{GOLDEN}(P_v, y_0 - \xi, y_0 + \xi, \epsilon)$ 
19: Output:  $[x_0^*, y_0^*]^T$ 
    
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V. NUMERICAL RESULTS

The simulations were conducted using a model representing a single user in a LoS free-space scenario, employing an elevated URA for localization. Table I summarizes the key parameters adopted for the Monte Carlo simulations. For benchmark comparisons involving the Nelder-Mead algorithm (e.g., as part of PEACH-MUSIC-NM), its parameters were configured as detailed in [1].

The performance of the PEACH-Golden method is evaluated against relevant benchmarks (as presented, for example, in Figure 1(a)), by varying the SNR. The Root Mean Square Error (RMSE), in meters, serves as the primary metric, quantifying the precision of the position estimates. Notably, even when its initial PEACH-MUSIC stage is configured for speed with a reduced search resolution (i.e., made "coarser"),

PEACH-Golden demonstrates the ability to achieve a localization variance comparable to that of a more finely-tuned PEACH-MUSIC baseline. However, PEACH-Golden does not attain the same level of ultimate accuracy as the PEACH-MUSIC-NM method; the latter benefits from leveraging the full URA pseudo-spectrum in its refinement stage, thereby operating closer to the CRB.

TABLE I
SYSTEM AND PROPOSED METHOD PARAMETERS

Parameters	Values
Realizations	$\mathcal{M} = 10^4$
Samples	$N = 10^2$
Architecture	8×8 , URA
Spacing	$\delta_x = 0.5\lambda, \delta_z = 0.5\lambda$
Carrier Frequency	$f_c = 15.0$ GHz
Transmitted signal power	$\mathcal{P} = 100$ mW
Path-loss Exponent	$\alpha = 2$
SNR	SNR = $[0 : 2 : 20]$ dB
Array Elevation	$z = 20$ m
User Position	$\mathbf{u} \sim \mathcal{U}[(−50, 50) \times (10, 50)]$ m
PEACH Parameters	
PEACH Search Region	$\Omega = [(-70, 70) \times (10, 70)]$ m
PEACH-NM	$N_{\text{NM}} = 15$
Hyperbola Candidates	$N_H = 96$
Circle Candidates	$N_C = 48$
PEACH-Golden Parameters	
Tolerance	$\epsilon = 10^{-5}$ m
Neighborhood	$\xi = 0.5$ m
Hyperbola Candidates	$N_H = 24$
Circle Candidates	$N_C = 12$

In Fig. 1(b), it is observed that in a scenario with no elevation, there is a high correlation between the URA's vertical antenna elements. This condition can hinder accurate user distance or height estimation, consequently increasing the RMSE. A drastic improvement in performance is evident at even small positive elevation angles, attributable to increased spatial diversity that benefits AoA-based localization techniques.

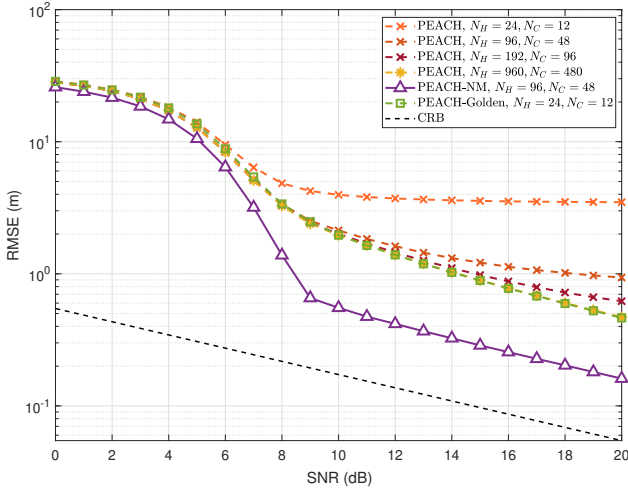
TABLE II
AVERAGED COMPARISON OF METHOD EXECUTION TIMES OVER PEACH

Method	$\overline{\text{RMSE}}_{10}$ (m)	$\overline{\text{RMSE}}_{20}$ (m)	$\overline{T}_m$ (ms)	$\overline{\mathcal{I}}_{\%}$
PEACH	2.14	0.92	1.10	0% (ref.)
PEACH-NM	0.55	0.16	1.10 + 5.65	513.36%
PEACH-Golden	1.99	0.47	0.47 + 0.44	-16.9%

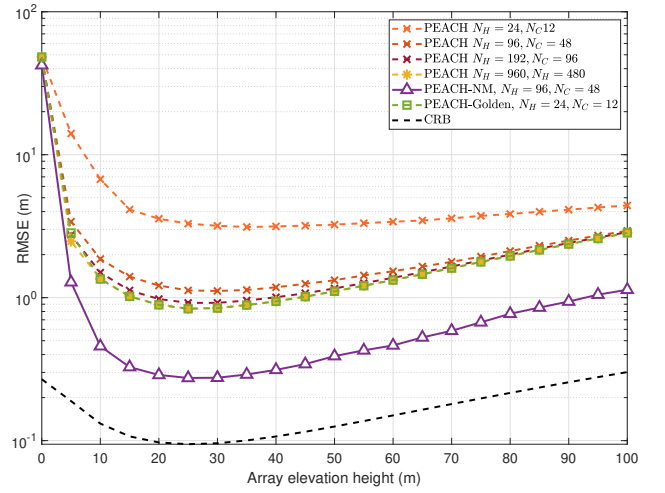
To rigorously assess the computational cost associated with each optimization technique applied to the PEACH-MUSIC algorithm, Table II presents a comparative analysis of their average execution times. The table quantifies how each variant performs in terms of processing duration, specifically highlighting the additional time and a percentage-based increase relative to the original PEACH method. The key metrics used for this comparison are defined as follows:

$\overline{\text{RMSE}}_n$: is the average position estimation error, measured in meters, corresponding to an SNR of n dB. This metric provides a direct measure of the methods' accuracy.

$\overline{T}_m$: is the method's average execution time in milliseconds (ms). This value indicates the total computational latency for



(a) RMSE evaluating the robustness of PEACH-based noise methods for different SNR, for different numbers of candidates.



(b) Effect of array elevation on localization based on intersection between circle and hyperbola.

Fig. 1. Comparison using the RMSE of PEACH-based methods as a function of SNR in dB and elevation height of the base relative to the ground of the array in meters. The results were obtained using different numbers of candidates for PEACH used in the circular and hyperbolic searches, respectively, to compare with the PEACH-Golden method, and the CRB.

each specific optimization approach evaluated.

$\bar{T}_{\%}$: is the average percentage increase in execution time relative to the baseline PEACH method. This normalized value provides a clear understanding of the relative computational burden imposed by each optimization.

While the PEACH-NM method exhibits the highest accuracy (as illustrated in Figure 1(a) and consistent with its near-CRB operation), Table II reveals that PEACH-NM incurs a much higher average execution time (T_m). Therefore, despite the inherent efficiency of the simplex algorithm in its refinement stage, the computational burden of utilizing the complete URA covariance matrix remains significant. In this context, PEACH-Golden particularly stands out by achieving a substantially shorter execution time than even the baseline PEACH-MUSIC method. This advantage in speed is rooted in the foundational principle of the PEACH framework: exploiting the URA's decomposition into orthogonal ULAs, which enables localization via the optimization of simpler, one-dimensional functions. Consequently, this approach necessitates computation of only the smaller covariance matrices associated with these individual ULAs, rather than the much larger full URA covariance matrix, leading to considerable computational savings.

VI. CONCLUSIONS

The PEACH-Golden method employs a two-stage approach for efficient localization. Initially, the PEACH-MUSIC algorithm, potentially configured for greater speed (e.g., using a coarser search), provides an initial estimate of the user's position. Subsequently, this estimate is precisely refined using a Golden Section search. Evaluation of numerical results (e.g., presented in Table II) reveals that PEACH-Golden achieves comparable performance and robustness to noise when compared to an optimal PEACH-MUSIC configuration, yet it drastically reduces the computational complexity. This efficiency is evident from its significantly shorter average

execution time, detailed in Table II. It is concluded that this method provides a substantial improvement over traditional PEACH-MUSIC in terms of computational load, making it particularly advantageous for localization in elevated-array scenarios (such as urban environments), where computational efficiency is often crucial. Furthermore, for applications requiring even higher precision localization, the PEACH-Golden framework can serve as a robust foundation for subsequent optimization stages or be integrated with other advanced techniques.

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